Massimiliano Baglioni

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Education

University of Perugia, B. in Computer Science – Perugia, IT — Oct 2019 – Apr 2023

- Graduated with 108/110.

University of Pisa, M. in Computer Science (Software: Programming, Principles, and Technologies) – Pisa, IT — Oct 2023 – Oct 2025

- Graduated with 110/110 with honors.

Open Source

Rust Clippy

Contributed to Rust Clippy, working on lint implementations, diagnostics, suggestions, and compiler-related code. View my contributions

Experience

Developer, Ds Digital Services – Assisi, PG, IT — Dec 2021 – Dec 2024

- Built dynamic websites and e-commerce platforms using WordPress as a CMS.
- Created and customized PHP plugins.

Developer, Cheapfit – Assisi, PG, IT — June 2022 – Sept 2024

- Built the website and e-commerce.
- Automated data analysis from the MySQL mobile application database.
- Implemented real-time user ranking with custom REST APIs in PHP.
- Developed frontend pages for live rankings using HTML, CSS, and JavaScript.
- Built an AI tool using OpenAI APIs with a PHP backend and React frontend.

Software Engineer, Advancia Technology – Viareggio, LU, IT — Nov 2025 – present

- Backend development using Java, Spring Boot, Spring Batch, OracleDB, MySQL, and Hibernate.
- Frontend development using Angular and RxJS.
- DevOps using Jenkins and Docker.

Projects

Bachelor Thesis Project (PHP / MySQL) — Jan 2023 – Apr 2024

Implementation of charts for student results analysis.

- Built a dynamic page for professors to analyze exam results using charts.
- Frontend implemented with Chart.js.
- Developed using MVC architecture with PHP and MySQL.

Master Thesis Project (Rust) Mar 2025 – Oct 2025

Fast Access through Lightweight Block-Based Compression Algorithms.

- Built a hybrid columnar-row store index in Rust.
- Implemented blocked bit-packing and lightweight compression to improve access time.

Compiler and Interpreter (OCaml) Jan 2025 – June 2025

Developed interpreters for MiniImp and MiniTyFun, and a compiler from MiniImp to MiniRisc with optimizations using OCaml.

- Implemented MiniImp, a subset of an imperative language.
- Developed MiniTyFun, a functional language with type checking.
- Built a compiler from MiniImp to MiniRisc (ILOC-style intermediate code) with optimizations.

Discord Music and Soundboard Bot (Rust) Oct 2024 – present

Discord bot to play audio from YouTube and Spotify links.

- Fully written in Rust, open source, and self-hostable.
- Plays soundboard with unlimited custom audio files.
- Actively used in multiple servers.

Skills

Programming: Proficient with Rust, Python, and Git; good understanding of Web, Linux, and Operating Systems concepts.

Other Programming Languages: Java, C, JavaScript, OCaml, PHP, SQL, Go.

Languages: English (fluent), Italian (native)